Section 4.2 — The Logarithm Function — Answer Key

Math 103: College Algebra

1 Translating Between Exponentials and Logs

Write each equation in logarithmic form.

1. $\log_5(625) = 4$
2. $\log_7\left(\frac{1}{2401}\right) = -4$
3. $\log_{3/4}\left(\frac{27}{64}\right) = 3$
4. $\log_1 3\left(\sqrt{13}\right) = \frac{1}{2}$

Write each equation in exponential form. Use your calculator to check that the new equation you have written is true.

5. $4^3 = 64$
6. $3^{-5} = \frac{1}{243}$
7. $\left(\frac{1}{5}\right)^{-2} = 25$
8. $25^{1/2} = 5$

2 Graphs of Logarithms

Identify each graph below as that of an exponential function, a logarithmic function, or neither.

9. Exponential
10. Logarithmic
11. Neither
12. Logarithmic

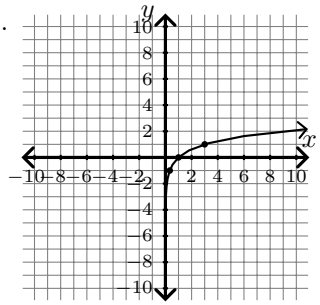
Of the two graphs above which are graphs of logarithm functions, choose the graph of each function below:

13. #10
14. #12

3 Graph a Logarithm

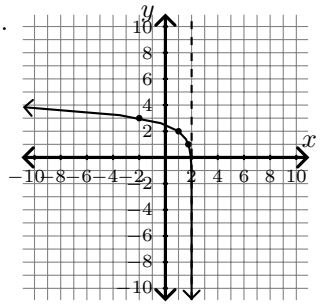
Sketch the graph of each logarithm function below in the space provided. Your graph should show the three points corresponding to outputs (not y values) -1, 0, and 1, as well as the asymptote. Do your best to estimate fractions.

15.



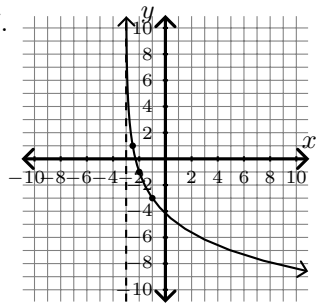
x	in	out	y
$\frac{1}{3}$	$\frac{1}{3}$	-1	-1
1	1	0	0
3	3	1	1

16.



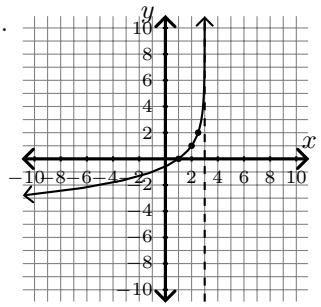
x	in	out	y
$\frac{7}{4}$	$\frac{1}{4}$	-1	1
1	1	0	2
-2	4	1	3

17.



x	in	out	y
-1	2	-1	-3
-2	1	0	-1
$-\frac{5}{2}$	$\frac{1}{2}$	1	1

18.



x	in	out	y
$\frac{5}{2}$	$\frac{1}{2}$	-1	2
2	1	0	1
$-\frac{5}{2}$	$\frac{1}{2}$	1	1

4 Domain and Range of a Logarithm Function

For each function below, give the domain and range.

19. $\operatorname{Dom} h = (3, \infty)$
    $\operatorname{Rng} h = (-\infty, \infty)$
20. $\operatorname{Dom} k = (-\infty, 5)$
    $\operatorname{Rng} k = (-\infty, \infty)$
21. $\operatorname{Dom} u = (3, \infty)$
    $\operatorname{Rng} u = (-\infty, \infty)$

5 Common Logarithm

For each log below, rewrite in exponential form, then determine whether the sentence is true.

22. $10^2 = 100$; true
23. $10^{-2} = 0.001$; false
24. $10^6 = 10000$; false
25. $10^{-5} = 0.00001$; true

Calculate each log on your calculator. Round your answer to two decimal places if necessary. Check by rewriting in exponential form.

26. $\log 12 \approx 1.08$
    $10^{1.08} \approx 12.02 \approx 12$
27. $\log 13,225 \approx 4.12$
    $10^{4.12} \approx 13182.57$
    ≈ 13225
28. $\log 0.03 \approx -1.52$
    $10^{-1.52} \approx 0.03$
29. $\log 1000 = 3$
    $10^3 = 1000$

6 Solving Equations Involving Logarithms — Basic

Solve each equation by rewriting in exponential notation. Round your answer to two decimal places, if necessary.

30. $x = 9$
31. $x = 16$
32. $x = 5$
33. $x = 20$
34. $x = 3$
35. $p = 1000$
36. $x = 1$
37. $x = 10^{1.1} \approx 12.59$